Evaluating predictive musculoskeletal simulations of walker-assisted gait in individuals with incomplete spinal cord injury

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Introduction

Walker-assisted gait is a common mobility strategy for people with incomplete spinal cord injury (iSCI), modifying gait mechanics by increasing upper limb support, promoting trunk flexion, and reducing speed. Understanding these adaptations supports rehabilitation and assistive technology design [1]. Predictive musculoskeletal simulation enables the development and evaluation of models to investigate how device characteristics, muscle-tendon properties, and control strategies affect healthy and pathological gait [2,3]. We developed a musculoskeletal model including a walker and evaluated whether it could reproduce gait adaptations associated with walker-assisted gait, providing a first step toward personalised predictive modelling.

Methods

Gait data were collected for two female subjects with different iSCI levels: severe bilateral -SCI 1, four rehabilitation sessions- and mild bilateral -SCI 2, two sessions- (Fig. 1a). Lower limb muscle strength was adjusted using the Manual Muscle Test scores [4]. A three-dimensional musculoskeletal model was defined per subject, including lower limb muscles and ideal torque actuators to control the arms [5]. A walker was incorporated, with both model hands rigidly connected to its handles. Contact spheres on the feet and walker defined ground interaction (Fig. 1b). Individual simulations were then performed for each subject and session in PredSim [2] using recorded gait speed.

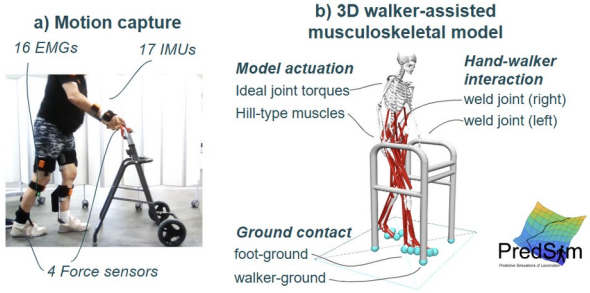


Figure 1: a) Distribution of systems used for motion capture of iSCI subjects. b) Outline of the used musculoskeletal model, with the implemented walker, displaying the muscles and contact geometry.

Results

Joint kinematics captured several trends, particularly hip movement, while knee ROM was generally overestimated. Ankle plantarflexion showed consistent longitudinal trends across rehabilitation sessions. Although the simulated changes were larger than those measured experimentally, the overall behaviour was preserved. In SCI 1, simulated maximum plantarflexion increased from 3.9° to 38.5° throughout the sessions, compared with 3.9° to 24.8° experimentally, whereas in SCI 2 it decreased from 12.6° to 8.1° in simulations and from 10.4° to 4.9° experimentally (Fig. 2b). Lower limb GRFs showed similar loading patterns, but walker forces were underestimated. Although the experimental measurements revealed different walker-use patterns between subjects, simulated walker GRFs remained at similar force profiles across rehabilitation sessions in both SCI 1 and SCI 2, indicating a limited ability to distinguish subject-specific walker reliance (Fig. 2c).

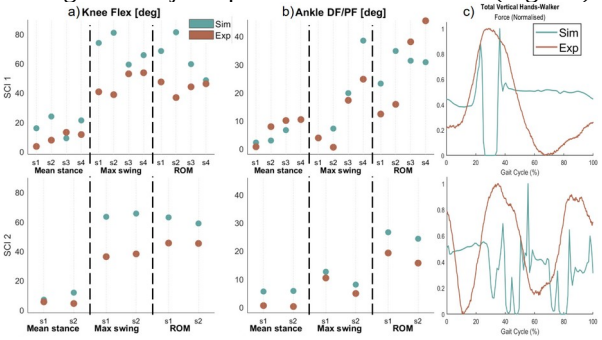


Figure 2: Comparison of simulated and experimental gait outcomes across rehabilitation sessions (s1, s2, …). a) Left column: Knee flexion metrics. b) Middle column: Ankle dorsiflexion/plantarflexion metrics. c) Right column: Total normalised vertical hands–walker force (representative session of each subject).

Conclusions

The musculoskeletal simulations reproduced the general characteristics of walker-assisted gait, but did not fully match the experimental measurements. These results support the potential of the model, while also highlighting the need for greater customization specific to each individual and their interaction with the walker. Ongoing efforts are focused on incorporating hand–walker interaction to improve the accuracy and subject-specificity of the simulations.

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References

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